Prompt 1-1: using the finite difference method

Solve this 2D plane stress problem from scratch using the finite difference method.

Problem details: A square domain $[0,1]\times[0,1]$ m with elastic modulus E=10e6 Pa, Poisson's ratio $\nu=0.2$, linear elasticity, and no body forces.

Boundary conditions:

- At y=0 (for all x): ux=0, uy=0
- At y=1, x∈[0,0.5]: ux=0, uy=0.1
- All other boundaries: traction-free

Calculations required: ux and uy displacement fields, strain fields, and stress fields.

Programming language: Python 3.10.

Permitted libraries: NumPy 2.1.1, SciPy 1.14.1, JAX 0.4.30, Matplotlib 3.9.2.